D-Separation as Certified Rewrite: Connecting Bayesian-Network Semantics to a World-Model Calculus

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Abstract

This note records a conceptual and formal "bridge" developed while mechanizing d-separation soundness in Lean and integrating it with a world-model calculus (WM-calculus) for probabilistic reasoning. The central message is simple: *d-separation is not merely a graph criterion for independence; it is a* ***certificate*** *that discharges side conditions of compiled probabilistic rewrites.* In the WM-calculus, inference rules are treated as Σ-guarded equalities on query semantics $e = \mathsf{ev}(W, q)$ rather than as globally valid manipulations of low-dimensional "truth values." The d-separation theorem becomes the canonical mechanism for turning structure into admissible rewrites.

We highlight three core accomplishments/ideas: (i) a decomposition of d-separation soundness into small, reusable lemmas (graph layer $\Rightarrow$ semantic layer), (ii) a discrete proof path that avoids heavy conditional-expectation machinery by reducing key identities to finite sums and rank-1 factorization on parent fibers, and (iii) a proof-architecture lesson: stateful invariants are essential to prevent branch explosion when converting between moral-ancestral trails and active trails.

A short philosophical interlude connects the "truth object + view" stance to empiricism and phenomenological interpretations of epistemic state, and notes a resonance with classical theories that treat inference rules as valid only under explicit preconditions. The note is intended as a primer for future work in the repository.

Status note. Implementation-status and count claims in this draft are snapshot-scoped; verify against current repository trackers and build outputs before citing as current state. Lean-proven claims are only those tied to explicit Lean file/theorem references; broader mathematical or historical claims are literature-backed context and not asserted as Lean-proven unless explicitly stated.

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1 Motivation: why d-separation matters for a WM-calculus

Many neuro-symbolic systems use fast local rules that look like "truth-value algebra" (e.g. update formulas on (strength, confidence) pairs). Such rules can be useful, but they are typically *not* globally valid without structural assumptions about independence and factorization. The WM-calculus stance is:

> *Make the semantic object primary: for a world-model state W and query q, define a truth object $e = \mathsf{ev}(W, q)$. Treat classical "truth values" as views/projections of e. Compile fast rules as Σ-guarded rewrites, where Σexplicitly states the assumptions under which the rewrite is exact.*

In this framing, d-separation is an unusually clean bridge: it turns *graph structure* into *semantic independence facts*, which in turn discharge Σfor many important probabilistic rewrites.

2 The WM-calculus interface (minimal form)

Definition 1 (World-model interface)**.** *A world-model class provides:*

1. *a type of states* State *(belief/posterior representation),*
2. *a type of queries* Query*,*
3. *a truth-object type* $\mathsf{Evidence}$*,*
4. *an evidence extraction map* $\mathsf{ev} : \mathsf{State} \times \mathsf{Query} \to \mathsf{Evidence}$*,*
5. *an update/revision operation* $\oplus$ *on states (possibly partial, to model overlap-aware fusion).*

Definition 2 (Σ-guarded rewrite)**.** *A compiled rule is represented as a theorem schema*

$$\Sigma \Rightarrow \mathsf{ev}(W, q) \equiv \mathsf{ev}(W, q').$$

Operationally, the system may apply the rewrite only when Σis discharged (proved) in the current context.

Remark 1 (Truth objects vs. views)**.** *A view is any map $v : \mathsf{Evidence} \to U$ into a tractable space U. The WM-calculus encourages using multiple views—for speed, control, or interpretability—without confusing them with the semantics.*

3 The d-separation bridge as a certified rewrite pipeline

This section records the conceptual pipeline that guided the mechanization.

3.1 Graph layer: active trails and moral-ancestral separation

Classically, d-separation is defined in terms of *active trails* in a DAG. The standard algorithmic equivalence states that d-separation can be checked by separation in an undirected graph:

Theorem 1 (Pearl-style equivalence (informal))**.** *Let G be a DAG, and let X, Y, Z be node sets. Then X and Y are d-separated by Z in G iff X and Y are separated by Z in the* moralized ancestral graph *of $X \cup Y \cup Z$.*

In a mechanized development, this equivalence is best proven by two explicit transforms:

1. **Forward transform:** an active trail $\Rightarrow$ an undirected moral-ancestral trail avoiding Z.
2. **Reverse transform:** a moral-ancestral trail avoiding $Z \Rightarrow$ an active trail.

Mechanization lesson. The reverse transform is where naive case splits tend to explode. A robust proof architecture uses a *stateful invariant* during trail-walking: "there exists some anchor node in X already actively connected to the current head, and the head is passable." This lets the proof proceed step-by-step along the moral trail, updating the anchor when a detour yields a new X node, and finishing immediately when a detour yields a Y node.

3.2 Semantic layer: factorization $\Rightarrow$ conditional independence

The semantic half of d-separation soundness is the global Markov property: if μ factorizes according to G (Bayesian network semantics), then d-separation entails conditional independence:

$$\mathsf{dSep}_G(X, Y \mid Z) \Rightarrow X \perp\!\!\!\perp Y \mid Z \quad \text{under } \mu.$$

For integration into a WM-calculus, the useful payload is not only the independence statement, but the algebraic identities it implies ("screening-off") that match compiled rewrite rules.

3.3 From conditional independence to rewrite equalities

A typical compiled rewrite needs an identity like

$$P(A, C, B)\, P(B) = P(A, B)\, P(C, B),$$

or equivalently $P(C \mid A, B) = P(C \mid B)$ under positivity assumptions. In the WM-calculus this becomes a certified rewrite:

$$\Sigma_{\text{d-sep}} \Rightarrow \mathsf{ev}(W, q_{\text{lhs}}) = \mathsf{ev}(W, q_{\text{rhs}}),$$

where $\Sigma_{\text{d-sep}}$ is discharged by a d-separation proof in the graph.

4 Major formalization breakthroughs (what actually made it work)

This section collects the key proof-techniques that turned "standard textbook math" into Lean-checkable theorems.

4.1 Breakthrough A: replace heavy `condExp` reasoning with discrete finite-sum algebra

A recurring obstacle in measure-theoretic formalizations is conditional expectation. In the discrete BN setting, the same identities can be proven *more robustly* by finite-sum computation:

1. Express $\mu(S)$ as a finite sum of joint weights:

$$\mu(S) = \sum_x \mathbf{1}_{x \in S}\, w(x)$$

   for measurable S.

2. Reduce screening-off on a parent fiber to a rank-1 factorization: on $\{\mathrm{Pa}(v) = c\}$, sums over descendants collapse by normalization (a telescoping argument), leaving dependence on x_v only through the local CPT term.

3. Conclude the cross-multiplication identity by algebra on sums.

Takeaway. In a discrete mechanization, it is often better to *prove the measure identity directly* than to force a large conditional-expectation lemma. The former is smaller, faster to elaborate, and aligns with the actual inference engine used operationally (variable elimination).

4.2 Breakthrough B: parent-fiber screening via rank-1 factorization

The key lemma shape is: for fixed parent assignment c and a set T measurable w.r.t. non-descendants excluding parents/self,

$$\mu(T \cap \{X_v = a\} \cap \{\mathrm{Pa}(v) = c\}) = \theta(a, c) \cdot K(T, c)$$

for some θ determined by the local CPT at v and some K independent of a. Once this is established:

$$\mu(T \cap \{X_v \in A\} \cap F)\, \mu(F) = \mu(\{X_v \in A\} \cap F)\, \mu(T \cap F)$$

follows by summing $\theta(a, c)$ over $a \in A$ and using $\sum_a \theta(a, c) = 1$.

4.3 Breakthrough C: stop branch explosion by carrying state in the reverse transform

The difficult direction in the moral-ancestral equivalence is:

$$\text{moral trail avoiding } Z \Rightarrow \text{active trail.}$$

A clean approach does *not* demand that each intermediate trail head lies in X. Instead it carries an invariant such as:

$$\exists x_0 \in X,\ \mathrm{HasActiveTrail}(x_0, \mathrm{head}) \quad \wedge \quad \mathrm{HeadPassable}(\mathrm{head}).$$

Then each moral edge step either:

- extends the state across a direct moral edge,
- passes through an activated collider witness (if the spouse witness is in $\mathrm{Anc}(Z)$),
- or triggers a detour lemma: "if the spouse witness is not in $\mathrm{Anc}(Z)$, then it reaches X or Y," which either finishes (hit Y) or updates the anchor (hit X).

Why this matters. This state-carrying style turns a combinatorial proof into a small number of reusable "step" lemmas and one recursion, which is exactly what proof assistants reward.

5 How this connects back to WM-calculus and PLN-style inference

5.1 D-separation as a Σ-discharge engine

The WM-calculus encourages treating fast rules as Σ-guarded rewrites. In practice, Σoften needs to express conditional independence.

Bayesian networks are the canonical class where independence is *structural*: it can be proven from the graph. Hence:

$$\Sigma_{\text{graph}} \Rightarrow \Sigma_{\text{semantic}} \Rightarrow \text{rewrite correctness.}$$

D-separation is the engine that supplies Σ_{graph}.

5.2 Truth values as views: why local propagation is incomplete

A recurring formal theme is that correlations live in the joint state. Therefore no fixed low-dimensional truth value (e.g. a pair (s, c)) can be globally complete across all queries. The WM-calculus response is to:

1. keep a truth object $e = \mathsf{ev}(W, q)$ that may be structured,
2. expose tractable views for control,
3. use certified Σ-guarded rewrites (e.g. from d-separation) when a view-level rule is exact.

5.3 Evidence overlap and partial fusion (why Σmatters even outside BNs)

When evidence overlap is possible, revision should be partial unless disjointness/independence is certified. This can be modeled by a separation-algebra-like structure for evidence, and by treating "combine" steps as admissible only under explicit preconditions. The same design discipline applies: *sound speed requires explicit* Σ.

6 Philosophical interlude: empiricism, phenomenology, and rule admissibility

The WM-calculus is compatible with a broadly empirical reading: the system does not claim access to "truth" as a metaphysical primitive; it maintains an epistemic state and extracts query-specific evidence objects. Views (like scalar probabilities or strength/confidence) are tools for action and communication.

Two resonances are worth noting:

- **Phenomenological stance:** a query is an intentional act; the "truth object" is the structured content made available under a particular perspective/intent (here, the current world-model state).
- **Rule admissibility tradition:** many classical principles are valid only under explicit preconditions (e.g. decidability, totality, or a chosen semantics). This parallels treating probabilistic rules as admissible rewrites only under Σ.

Classical epistemological systems that foreground *conditions of valid inference* align well with the WM-calculus discipline: inference is licensed by explicit warrants rather than assumed universally.

7 Outline of the mechanized pipeline (as a roadmap)

For future work (or new contributors), the mechanization pipeline is:

1. **Graph:** formalize active trails and d-separation; formalize moralization/ancestral construction.
2. **Equivalence:** prove the moral-ancestral equivalence by forward and reverse transforms using stateful invariants.
3. **Semantics:** define discrete CPT semantics; expose μ as a measure and $w(x)$ as joint weights.
4. **Local Markov:** prove screening-off on parent fibers by finite sums and telescoping elimination.
5. **Global Markov:** compose graph equivalence with semantic CI to obtain d-separation soundness.
6. **WM integration:** expose d-separation proofs as Σcertificates discharging rewrite rules.

8 Relationship to companion repository notes

This note intentionally focuses on the d-separation bridge and the mechanization lessons that make it tractable. Companion notes cover broader or orthogonal topics:

- a general WM-calculus primer (truth objects, views, evidence algebras),
- algorithmic probability and Solomonoff-style induction notes,
- process-calculus formalization notes,
- bootstrapping neurosymbolic agent execution/inference-control notes.

9 Conclusion

The d-separation theorem is more than a classical result in graphical models: it is a reusable *certificate generator* for Σ-guarded probabilistic rewrites. Mechanizing it clarifies where independence assumptions enter, how they can be discharged structurally, and how to keep fast inference rules honest inside a world-model calculus.

References

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